What are APIs ?

By
Zafar Iqbal

API Call

API Call

Cake = Bakery7.GetCake(2 pound, 3000)

Endpoint

Function

Subscription Key

Karahi = Restaurant4.GetKarahi(2kg, 8000)

Endpoint

Function

Subscription Key

Purpose of API Call

- API allows different software applications to communicate and share data, enabling them to work together.

- APIs simplify software development by enabling developers to integrate data, services, or capabilities from other applications, rather than needing to develop them from scratch.

API Call --- Components

API client:

API client makes it easier for a person to use an API while hiding the complexities of the API backend details.

Endpoint:

An endpoint is a dedicated Uniform Resource Locator (URL) that points to the location of a resource on a server.

API key:

A unique passcode containing letters and numbers that grants access to an API. These keys often authenticate an application or identify an authorized user.

API request:

API request is a message sent to an application asking a server for information or a service.

API Call -- Format

HTTP Method

GET

https://socialmedia.com/v1beta/GetComments?User=UserId&apikey=AIzaSyB14KbVfFYxUwoF0U6OpcenAJY-i5JVOKE

Endpoint

Parameter

API Key

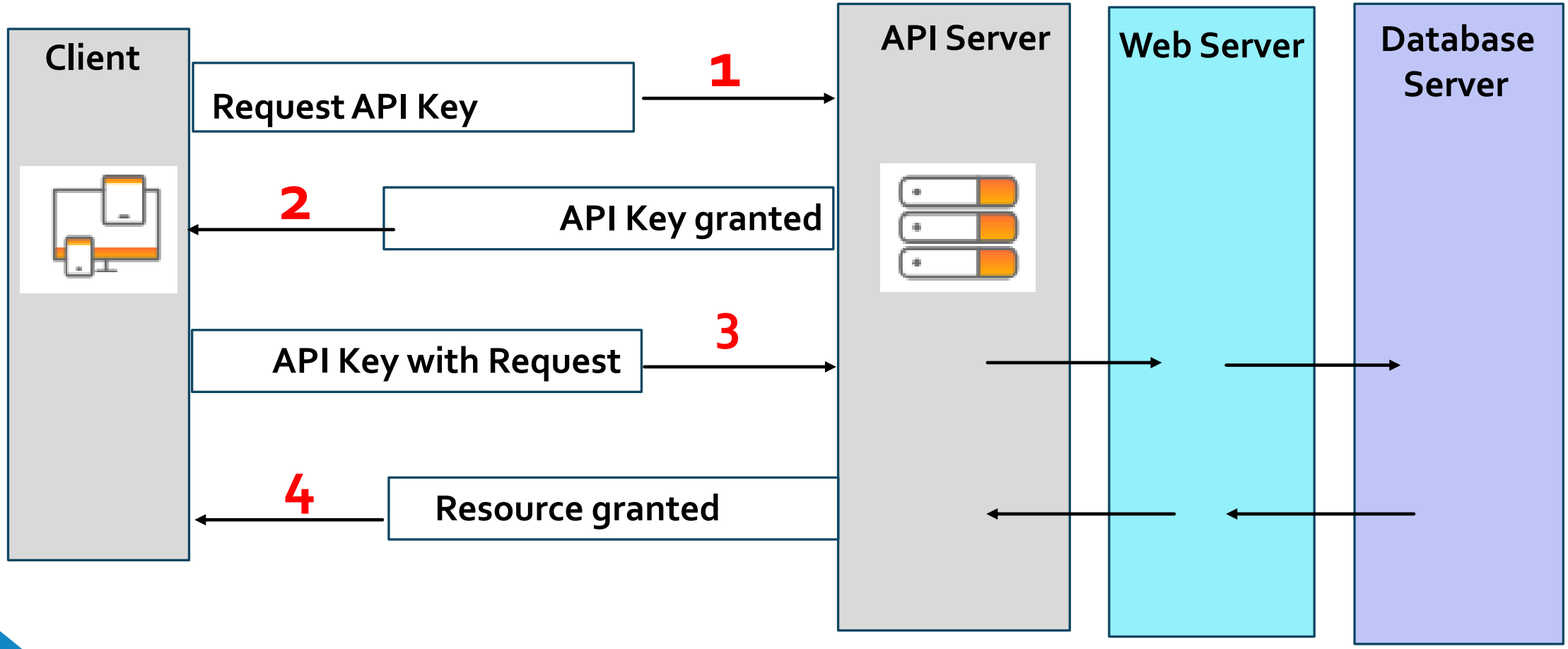
POST

https://sociamedia.com/v1beta/SubmitComment?User=UserId&apikey=AIzaSyB14KbVfFYxUwoF0U6OpcenAJY-i5JVOKE

```
{"Comment":"This looks excellent.",
  "rating" : "5/5"
}
```

Request Body

Client - API Server communication

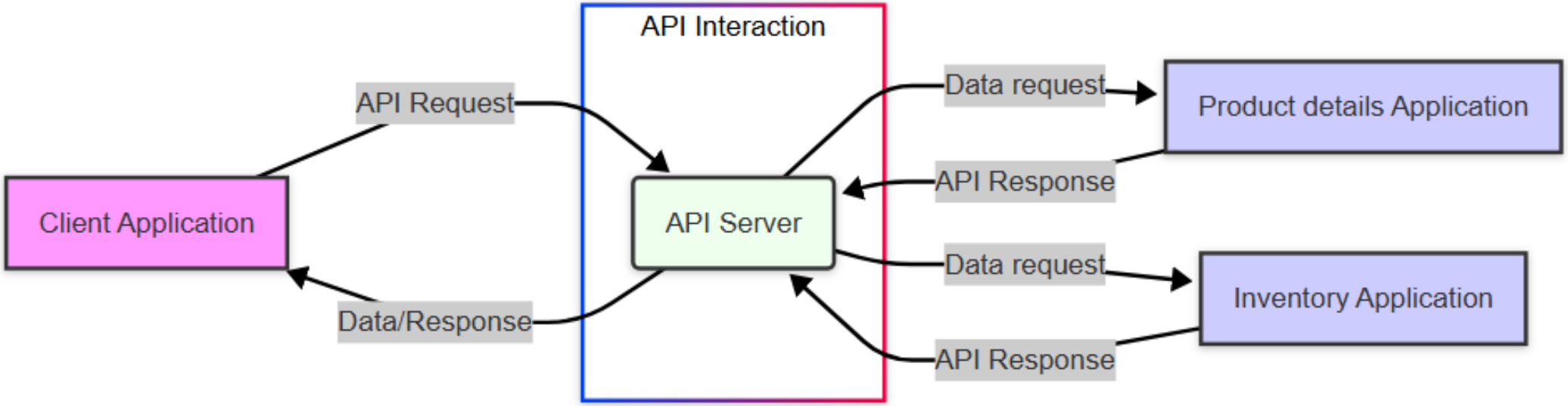


Web Application & API

Web Application:

An online shopping website where users browse, select, and purchase products. For example, Amazon

API: The backend service that allows web application (Amazon) to retrieve product information, check inventory or process payments.



Advantages of APIs in AI/ML Development

In AI development, APIs enable AI developers to integrate sophisticated AI functionalities into applications without building complex AI models from scratch, streamlining development and making advanced technologies accessible.

Advantages:

- **Access to AI Capabilities:** APIs help Integrate AI functionalities from services like OpenAI, Google Cloud, or Microsoft Azure into their applications.
- **Reduced Development Time:** Instead of building AI models from the ground up, developers can leverage pre-trained models and algorithms through APIs, significantly reducing development time and effort.
- **Escape from AI Complexities:** Developers can concentrate on the required features instead of getting Involved in the complexities of AI implementation.
- **Reduced costs:** By leveraging pre-built models and algorithms, developers can reduce development costs.

Difference between web application & API

Feature	Web Applications	APIs (Application Programming Interfaces)
Definition	A software application accessed via a web browser	A set of rules that allow different software systems to communicate
Purpose	Provides user interaction and visual content	Facilitates data exchange and functionality integration
User Interface	Has a graphical UI for users	Typically lacks a UI, responds with data
Access	Accessed via browsers (Chrome, Firefox, etc.)	Accessed programmatically via requests (HTTP, REST, GraphQL, etc.)
Data Format	Presents data in HTML, CSS, and JavaScript	Returns data in JSON, XML, or other structured formats
Interactivity	User-driven interaction through forms, buttons, etc.	Machine-to-machine communication with requests and responses
Usage	Used by end-users	Used by developers and software applications
Examples	Facebook, Gmail, Amazon	Google Maps API, Twitter API, PayPal API

Examples of AI APIs

- **OpenAI API:**

Provides access to powerful language models for tasks like text generation, translation, and code generation.

- **Google Cloud AI APIs:**

Offers a suite of machine learning and AI tools for vision, speech, natural language, and structured data analysis.

- **Microsoft Azure AI Services:**

Provides pre-built machine learning models and algorithms for tasks like emotion detection, facial recognition, and speech recognition.

- **IBM Watson:** An AI platform that helps developers create intelligent apps with natural language understanding and big data analysis capabilities.

- **Clarifai:** Specializes in visual recognition and computer vision technology.

Aassignment

Retrieve API key for Google Gemini API.
What is API Authentication ?